

Capstone Project Specification

AI Engineering Cohort

Objective

This capstone requires participants to validate their problem definition, architecture, and evaluation methodology before implementation. Participants must avoid common pitfalls like vague objectives or lack of objective performance metrics.

Design Doc Requirement

Approval of the design doc is a prerequisite for receiving an OpenAI API key.

Required Design Doc Sections

1. **Problem Statement:** A specific, narrow definition of the challenge being addressed.
2. **Architecture:** A high-level overview of the system design (RAG, ReAct, etc.).
3. **Evaluation Criteria:** Specific quantitative and qualitative success metrics.
4. **Framework Justification:** Selection of frameworks (e.g., Pydantic AI, smolagents, LangGraph) with reasoning.

Group Allocation:  Capstone Groups

One Pager Submission: [Submission Form](#)

Project Requirements

Functional systems must include:

- **Data Surface:** A knowledge base or tool-use interface that the system actively queries.
- **Core Flow:** An implementation of a retrieval or orchestration flow, such as a RAG pipeline, a ReAct-style loop, or Multi-Agents System.
- **Comparative Evaluation:** A mandatory evaluation comparing at least two distinct approaches (e.g., Vanilla RAG vs. Graph RAG vs. Agent), with a justified final choice.
- **Failure Analysis:** A README section documenting attempted approaches, failures, and subsequent pivots.
- **User Interface (Optional):** A functional interface that allows external users to interact with the system.

Project Timeline

Milestone	Deadline
One-Pager Submission	Tuesday, Sep 15, 2026
Build & Evaluation Phase	Sep 16th - Sep 26th, 2026
Project Submission Deadline	Sunday, Sep 27th, 2026
Mentor Review Period	Sep 27th – Oct 3rd, 2026

Deliverables

- **Design Doc:** 1–2 pages.
- **Code Repository:** Access must be provided for review (waived for proprietary corporate projects).
- **Documentation:** max 4–5 pages.
- **Demo Video:** A concise 3-minute walkthrough of the system.

Evaluation Rubric

Criteria	Focus Area
Problem Specificity	Is the problem statement narrow and well-defined?
Data Processing	Effective Data Ingestion Pipeline
Retrieval Design	For RAG or search optimization.
Orchestration	Logical design of single/multi-agent tool use, Handling Context, etc
Evaluations	Use of LLM-as-a-judge and quantitative metrics.
Decision Reasoning	Justification for architectural and framework choices.

Note: Reused projects or fabricated evaluation data will result in a score of zero.

Common Failure Modes to Avoid

- **Vague Problem Statements:** "An AI that helps with everything" is not a project.
- **Architecture-First Bias:** Selecting complex frameworks before establishing a baseline or understanding data.
- **Complexity Bias:** Example, Skipping a simple baseline before adding advanced retrieval techniques.

For project inspiration, participants are encouraged to review the [Y Combinator Requests for Startups](#).